

Reflection on Sprint 0, Week 3

Seb Rusbridge, Oct 2026

Our group attempted to learn how to programme linear regression modeling and some machine learning techniques and evaluate how we could use them on our education dataset to predict a student's final grade using demographic, social, behavioral, and educational characteristics. Starting with the notes, we then researched and found articles on how to use python to fit linear models (I already had experience doing this on R but wanted to try and learn how to do this on python instead), and create random forests and gradient boosted regression models. Using the "Student Performance in Secondary Education" dataset, we then applied these techniques to see if we could perform a relatively basic analysis of the data.

In our initial exploratory modeling, stepwise selection ($p < 0.05$) selected G2 (second-period grade), G1 (first-period grade), number of past class failures, and school selection reason should be chosen as covariates for the model, subsequently achieving an R^2 of 0.851 when we fitted the linear model. We also found the same pattern with random forests and GBR's. This allowed us to fit a pretty good linear model on certain covariates. However, these covariates were not the ones we wanted to analyse from the dataset. This meant we had to engineer the dataset a bit in order to get more meaningful results. So, we intentionally excluded G1, G2, and failures going forwards. However, once omitted, the results we got suggested that previous academic performance may contain important information for predicting final grades that was not fully captured by the remaining student characteristics included in the dataset.

Another challenge for me, specifically, was translating knowledge of programming linear regressions in R into Python. I understood the basic statistical method, but I found that this didn't mean I could easily implement it in a different programming language. I had to become familiar with packages such as statsmodels and learn more about how Python handles dataframes and model fitting. In future projects, I would like to become more comfortable with the Python data-science ecosystem instead of just trying to repeat techniques from R.

This also highlighted some limitations in our literature search. We began with the lecture notes and then searched for some examples of implementing the different models in Python online. This was useful, but our search wasn't very systematic. Kaggle examples were useful for seeing how code might need to be structured, but it often focused on obtaining results more than critically evaluating them, when we wanted to try and do both.

Linear regression was the part where I felt most comfortable mathematically at the start. I understood the basic idea of estimating coefficients by minimising the residual sum of squares and could interpret the resulting coefficients. I also knew about the assumptions required for the model to be appropriate and how to identify if one of these assumptions had been violated, though I did not know how to produce the graphs used for diagnosis on python initially, and I knew how we could remedy when these assumptions had been violated using techniques like

transformations. I was also less familiar with what actually happens inside Random Forests and Gradient Boosting at the start. Although together, we could implement these models, I would like to understand the actual processes they use better so that I can make better decisions about when to use them instead of just looking at their outputs.

The project also exposed a number of gaps in my knowledge of collaborative coding, specifically on GitHub. We spent a significant portion of our time getting GitHub and Notable to work as intended, and, for a while, I was unsure about if and when to use SSH keys or personal access tokens and what they actually do. We also had to learn how forking and pull requests worked, and why they might be important. Sometimes I would follow instructions to solve a problem I was having on GitHub without fully understanding why the solution worked, though this happened less towards the end of the project. This is an area in which gaining additional knowledge would make me feel much more confident. Understanding commits, forks, pull requests and authentication earlier on would have allowed me to work better collaboratively and reduce the large amount of time lost to technical issues.

Overall, the project showed me that data science involves more than simply fitting models. I could benefit from developing greater fluency with statistics in Python and improve my understanding of GitHub. I would also like to strengthen my mathematical understanding of machine-learning methods further, as we have recently been doing in lectures, so that I can better understand not just how to programme them, but when and why they should be used.